MIND SPRINT – Hiring Opportunities Module

Interview Explanation & Technical Documentation

1. Module Overview

This module is part of the final year project 'MIND SPRINT – AI Career Companion for BCA Students'. The Hiring Opportunities module helps BCA students track top off-campus hiring companies, view live hiring updates, understand selection processes, and save companies of interest through a single interactive dashboard.

2. Problem Statement

Freshers often struggle to track off-campus hiring because information is scattered across multiple websites. Deadlines are missed and there is no centralized platform focused specifically on BCA students. This module solves that problem by aggregating reliable hiring data with live updates.

3. Key Features

1 Dashboard displaying top off-campus hiring companies

2 Live hiring updates using web scraping

3 Search and filter functionality

4 Save/bookmark companies for later reference

5 Detailed company view including roles, selection stages, and syllabus

6 Hiring trends and preparation tips for BCA students

4. Working of the Module

1 UI Initialization using Streamlit with custom CSS styling

2 Static company data stored in structured Python dictionaries

3 Live hiring data fetched using web scraping techniques

4 Refresh mechanism to update hiring information in real time

5 Session state management to store saved companies and user interactions

6 User interaction flow from search to application redirection

5. Technologies Used

1 Frontend: Streamlit, HTML, CSS

2 Backend Logic: Python

3 Web Scraping: Requests, BeautifulSoup, Regular Expressions

4 State Management: Streamlit Session State

5 Tools: VS Code / PyCharm, Git, GitHub

6. Challenges & Solutions

1. Website structure changes – solved using fallback static data and keyword-based scraping
2. Performance during scraping – solved using controlled refresh and progress indicators
3. No database usage – solved using efficient session state management

7. Learning Outcomes

This module helped in gaining hands-on experience with real-world web scraping, dashboard UI design, session management, and building scalable Python applications.

8. Interview Closing Statement

This module demonstrates my ability to combine Python, web scraping, and frontend development to solve a real-world problem faced by freshers.